Interlude: Bionumbers and Project

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

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1 Interlude 1: Bionumbers & Final Project

We've emphasized rules of thumb – numbers you can remember to make a lot of different calculations. Example: 1 molecule in a typical bacterial volume ($1\ \mu\mathrm{m}^3$) has a concentration of 1 nM $= 10^9$ M.

But you can't remember everything. For this we have *BioNumbers* – **a compendium of biological numbers with units references and notes**.

Today we'll do a tour of BioNumbers (and related resources) and discuss your final projects.

Links to peruse

1. BioNumbers examples [BNID 116987, 103488, 104070, 101759]
2. Snapshot: Key Numbers in Biology
3. Snapshot: Timescales in Cell Biology
4. The human brain in numbers: a linearly scaled-up primate brain
5. The Anthropocene by the Numbers

Outline for whiteboard

- HW
- finish lecture #4
- groups
- bionumbers tour
- snapshots
- big insects
- bionumbers exploration
- project intro
- project chatter
- teaser – diffusion limited enzyme (?)

2 Example: huge insects in the carboniferous

The Carboniferous period spans 60 million years, from the end of the Devonian 358.86 Ma (million years ago) to the beginning of the Permian 298.9 Ma. For various reasons we think that the Carboniferous atmosphere contained about 50% more O_2 than today, meaning the O_2 partial pressure was $pO_2 \approx 30\%$

One of the reasons for this is that we have fossils of really big insects from this period, fossils like Meganeura – a dragonfly with a 75 cm wingspan (Fig. 1) – and Arthropleura – a giant millipede that reached lengths of at least 2 m and left tracks at least 0.5 m wide (Fig. 2).

What's the connection between big insects and pO_2?

The idea is that insects have open circulatory systems (Fig. 4). Quoting from Encyclopedia Britannica:

> The [insect] respiratory system consists of air-filled tubes or tracheae, which open at the surface of the thorax and abdomen through paired spiracles. The muscular valves of the spiracles, closed most of the time, open only to allow the uptake of oxygen and the escape of carbon dioxide.

So the insects are, in some sense, a hollow tube of air. They need oxygen to diffuse from the hollow organs through their tissues. So more O_2 in air means (i) a larger concentration difference between air and tissue and, therefore, (ii) a larger diffusive flux $J = -D\frac{dO_2}{dx}$ that sustains a larger tissue, as in Fig. 3.

2.1 The question

A modern dragonfly has a body width of a $w = f$ mm. The carboniferous dragonfly Meganeura had a body width of a $w = f$ cm. So it was about tenfold wider. **Could a 50% increase in O_2 plausibly sustain a tenfold wider insect?**

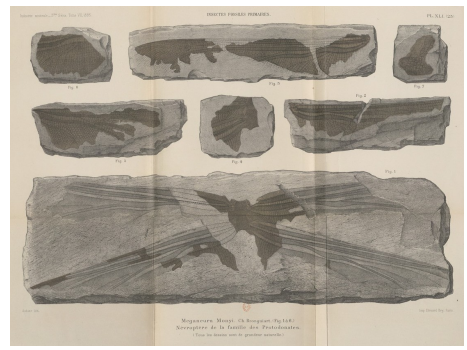

Figure 1: An illustration of a Meganeura fossil by Brongniart, 1894. Via Wikipedia. These extinct Carboniferous dragonflies had wingspans of 75 cm (0.75 m).

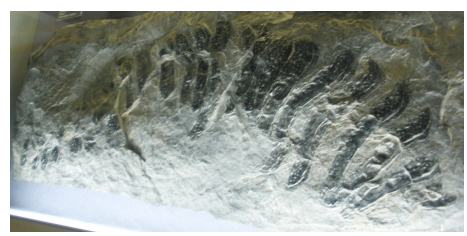

Figure 2: The "Maybach specimen" of Arthropleura armata, a 90-centimetre (35 in) long specimen missing the head and telson. Via Wikipedia.

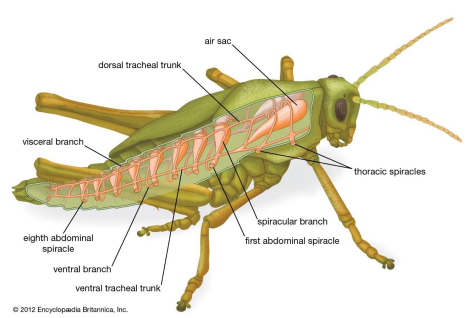

Figure 4: The open respiratory system of a grasshopper from Encyclopedia Britannica Wikipedia.

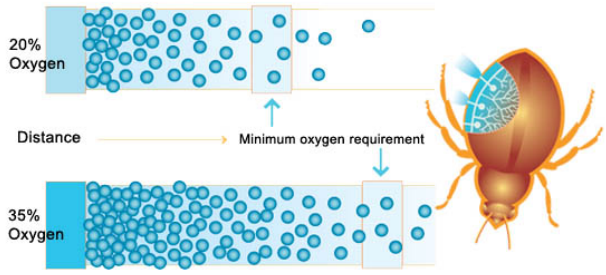

Figure 3: A simplified diagram of the hypothesis, which is summarized as "more oxygen $\implies$ larger insects." Diagram via ASU.

2.2 The calculation strategy

We know modern dragonflies thrive in 20% O_2, which means that diffusion can supply their tissues with enough O_2. We'll assume

1. $\dot{q} \approx 3$ W/kg metabolic rate like today's insects [BNID 107981].
2. a typical metabolic energy of O_2 respiration $H \approx 500$ kJ/mol[1]
3. a tissue density of $\rho \approx 1$ gm/mL for ancient and modern.
4. an diffusion constant for O_2 of $D_{O_2} \approx 10^3$ $\mu\mathrm{m}^2\,\mathrm{s}^{-1}$

This sets up a metabolic balance where

$$J_{diff}(L) = J_{met}(L).$$

That is, diffusive supply of O_2 must balance metabolic consumption and both of them are functions of size $L = w/2$.

2.3 The calculation

We estimate the diffusive flux across the tissue using Fick's first law

$$J_{diff} = -D\frac{dO_2}{dx} \approx -D\frac{\Delta O_2}{L} \approx -D\frac{pO_2}{L}.$$

Here we are treating the whole tissue as a layer and assuming that most of the O_2 is depleted at the end, so that $\Delta O_2 \approx pO_2$ in the appropriate units. From here on we will drop the minus sign because we are balancing absolute fluxes and don't care about their orientation in space. This flux has dimensions of [# L^{-2} T^{-1}] and the relevant area is a cross-section of the body along its long dimension.

What about J_{met}? We assumed a constant power consumption per unit volume of $\dot{q} \cdot \rho$. This expression has units of W/m^3 = J/s/m^3, so if we divide by the molar energy content of O_2, we can get an expression $\dot{q} \cdot \rho / H$ in units of mol O_2/s/m^3. If we multiply this expression by the appropriate length scale ($L = w/2$), then we get dimensions [# L^{-2} T^{-1}] matching the diffusive supply.

This gives $\frac{\dot{q}}{H}\rho L$ where $\frac{\dot{q}}{H}$ is O_2 consumption per unit mass and ρL convert it to O_2 consumption per unit cross-sectional area.

So our balance is

$$D\frac{pO_2}{L} \approx \frac{\dot{q}}{H}\rho L \implies L^2 \approx \frac{H \cdot D \cdot pO_2}{\dot{q} \cdot \rho}$$

[1] Glucose respiration has a reaction free energy of ≈ 3000 kJ/mol and consumes 6 O_2 per glucose.

and

$$L_{max} \propto \sqrt{pO_2}$$

If pO_2 was 50% higher in the Carboniferous

$$\frac{L'_{max}}{L_{max}} = \left(\frac{pO'_2}{pO_2}\right)^{1/2} = \sqrt{1.5} \approx 1.2$$

So we can justify a 20% increase in body width with a diffusive argument, but definitely not a tenfold increase.

2.4 Some comments and issues with this calculation

Notice that **we didn't actually to calculate the metabolic rate**. Nor did our estimate require a concrete value of D. By taking the ratio we were able to cancel out everything we assumed was constant, namely D, $\frac{\dot{q}}{H}$, and ρ.

There is a wrinkle here. We assumed that metabolic rates scale linearly with mass so that an insect with twice the mass has double the metabolic rate. It turns out that **metabolic scaling is not linear but sub-linear** with $\dot{q} \propto M^{\alpha}$ with $\alpha = 0.6 - 0.8$ depending on the type of organism and the group doing the analysis. You can read about this in the vignette titled "How does metabolic rate scale with size?"

Should we account for sub-linear scaling? It does mean that larger insects respire less (per unit mass) so that larger insects have somewhat lower mass-specific oxygen demand $\frac{\dot{q}}{H}$. **Could this sub-linear scaling won't affect our conclusion?**

Last but definitely not least: there is a geometric issue. The insect is not really a hollow tube. It's respiratory system has a complex geometry (Fig. 4) that affects the surface area available for diffusion. A detailed calculation would be needed to think about those effects.

2.5 Alternative explanations

If O_2 supply fails to explain tenfold larger insects in the carboniferous, what are some potential explanations?

One possibility is that we failed to consider the right metabolic rate. We considered a resting metabolic rate, which we assumed to be proportional to mass. It's possible that some other metabolic rate, e.g., during flight, has a very different relationship with body size and/or environmental temperature. This is an avenue explored in recent literature, e.g., Verberk & Bilton DT. PLoS One. 2011.

Another alternative is predation, or lack thereof. Birds are major modern predators of insects and birds are dinosaurs. As such, birds did not arise til the Jurassic, which is ≈ 100 million years after the Carboniferous.

Another possibility is that larger organisms are more likely to leave fossils, which would skew our appreciation of the sizes of insect bodies throughout history. This is called "taphonomic bias."

Can you think of any other potential alternative explanations? Some options are discussed in Harrison et al. Proc Biol Sci. 2010.

2.6 Bionumbers we'd like to have

This calculation highlighted some useful BioNumbers entries (cited above) and some that are missing or sparse.

Missing entries include

1. morphology of modern and fossil insects
2. especially of their respiratory systems
3. insect metabolic rates and scaling as in Chown et al. Func. Eco 2007.
4. diffusion of oxygen in tissues, e.g. [BNID 115120]

Can you think of others?

2.7 Relevant references

1. **Graham JB, Aguilar NM, Dudley R, Gans C. Implications of the late Palaeozoic oxygen pulse for physiology and evolution. Nature. 1995;375(6527):117-120. doi:10.1038/375117a0
2. **Snelling EP, Lensink AV, Clusella-Trullas S, et al. Oxygen supply through the tracheolar-muscle system does not constrain insect gigantism. Nature. Published online March 25, 2026:1-5. doi:10.1038/s41586-026-10291-3
3. **Raff RA, Raff EC. Respiratory mechanisms and the metazoan fossil record. Nature. 1970;228(5275):1003-1005. doi:10.1038/2281003a0
4. Chown SL, Marais E, Terblanche JS, Klok CJ, Lighton JRB, Blackburn TM. Scaling of insect metabolic rate is inconsistent with the nutrient supply network model. Funct Ecol. 2007;21(2):282-290. doi:10.1111/j.1365-2435.2007.01245.x
5. Verberk WCEP, Bilton DT. Can oxygen set thermal limits in an insect and drive gigantism? PLoS One. 2011;6(7):e22610. doi:10.1371/journal.pone.0022610
6. Harrison JF, Kaiser A, VandenBrooks JM. Atmospheric oxygen level and the evolution of insect body size. Proc Biol Sci. 2010;277(1690):1937-1946. doi:10.1098/rspb.2010.0001

3 BioNumbers group work

Get to know your co-mentor. Talk about numbers that would be useful in your current or prior research. Find one useful BNID and one that is missing. Write them down. Take ≈ 20 min to work together.

Huge kudos if you make an account and submit a BioNumber during class.

4 Project intro and time

Your project consists of two parts: a whiteboard presentation (5 min) and a written vignette (2-3 pg) that summarizes your project and the calculations you did, together with any other relevant details.

You should **prepare for these 5 minutes of glory** – practice seriously with the help of classmates. We encourage and will give bonuses for including additional insights in your vignettes, e.g., lessons, calculations, or sanity checks beyond what you show in the presentation.

Class presentations should be 5 minutes long. If you have a creative idea for presenting the topic or the calculation (e.g. involving the audience, bringing a demonstration, etc.) you are entitled to two extra minutes.

We know that you already set aside MW 10-11:30 for this class, so we have scheduled some class time to work on the project together. We'd like an initial project idea by e-mail by 05/04/26. Additional timeline information is in Table 1

When	What to Do
05/04/26	Send initial project idea to course staff
2+ weeks before talk	Project idea approved by course staff
1+ week before talk	(i) practice with co-mentor on whiteboard (ii) e-mail us a paragraph about your practice. how it went and a useful suggestion you got
during other's presentations	(i) come to all presentations, listen actively (ii) following each session, send a comment to one presenter (details in the text) if you can't attend, you will need to read and comment on one vignette per session
<u>1 week</u> after your presentation	Upload your final vignette. Include any feedback you collected.

Table 1: Final Project Presentation Requirements

You are expected to attend all final presentation sessions and actively listen to the talks by your fellow students. Following each session, choose one topic to briefly comment on. The comment should be an idea for an interesting follow-up, a suggested correction, improvement for presentation style etc. The comment should be emailed to the presenting student, with the course email CC'd.